PRISM: A Multi-Agent Forensic Verification Framework Using Dual-Stream Transformers, Sensor Noise Analysis, and Evidential Reasoning

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Abstract— *Deepfakes produced by diffusion models, generative adversarial networks (GANs), and neural vocoders defeat monolithic binary classifiers, which lose 20%–35% accuracy on unseen generators or heavily compressed media and offer neither spatial explanation nor legal auditability. This paper presents PRISM (Provenance and Reasoning Intelligence System for Multimedia), a modular multi-agent framework for forensic verification. PRISM couples a dual-stream backbone—consisting of a Vision Transformer (ViT-B/16) adapted with weight-decomposed low-rank adaptation (DoRA) and a photo-response non-uniformity (PRNU) sensor-noise branch—with eight forensic agents that examine container metadata, compression error levels, optical-flow kinematics, three-dimensional illumination, chrominance-based remote photoplethysmography (rPPG), corneal reflections, lip-sync timing, and vocoder spectra. Agent outputs are fused by Dempster–Shafer evidential reasoning, which treats a missing modality as epistemic uncertainty rather than as evidence for or against tampering. PRISM produces a 224x224 tampering-localization mask and a structured report aligned with NIST SP 800-86. Evaluated on FaceForensics++, Celeb-DF v2, DF-Platter, and ASVspoof 2021, it reaches 95.00% accuracy, 0.9740 AUC-ROC, 3.80% equal error rate, and 0.7840 localization mIoU. On a consumer NVIDIA RTX 3050 laptop GPU (6 GB VRAM), it runs at 23.9 frames per second (41.9 ms latency) with 1,450 MB of GPU memory.*

IEEE Keywords— Deepfake detection, Dempster–Shafer theory, digital forensics, error level analysis, low-rank adaptation, multimedia forensics, photo-response non-uniformity, remote photoplethysmography, Vision Transformer.

I. INTRODUCTION

GENERATIVE artificial intelligence has sharply lowered the cost of producing photo-realistic media forgeries. Latent diffusion models [1], GANs [2], and neural vocoders [3] now enable face replacement, expression reenactment, and voice cloning at scale. When such content circulates through social media and encrypted messengers (e.g., WhatsApp and Telegram), aggressive lossy re-encoding strips container provenance and masks subtle generative artifacts.

Current forensic toolchains have three fundamental weaknesses in real-world deployment:

1) Poor cross-dataset generalization: Monolithic classifiers trained on constrained benchmarks such as FaceForensics++ [4] overfit to dataset-specific rendering footprints. On unseen generators or camera sensors, accuracy commonly degrades by 20%–35% [5].

2) No spatial explanation or audit trail: A black-box network returns a scalar forgery score without indicating which regions or frames are anomalous. Under NIST SP 800-86 [6] and SWGDE guidance [32], an unexplained score is weak evidence without corroborating physical findings.

3) High computational cost: State-of-the-art detectors typically need 16–24 GB of GPU memory, which field examiners and local law-enforcement units rarely have.

To address these gaps, we present PRISM (Provenance and Reasoning Intelligence System for Multimedia). Analogous to clinical differential diagnosis, PRISM examines media along eight complementary physical, biological, and spectral dimensions and fuses the findings with Dempster–Shafer theory [7], so that absent modalities are handled explicitly. The main contributions are:

- A dual-stream backbone combining a DoRA-adapted ViT-B/16 with a PRNU noise branch through cross-attention, training only 1.2M parameters.
- Eight specialized forensic agents, each emitting an interpretable evidential mass with a stated rationale.
- An evidential fusion engine that separates 'no evidence' from 'evidence of authenticity,' flags inter-agent conflict, and preserves decisive tampering findings.
- A pixel-level localization decoder and an audit-ready report with cryptographic chain of custody.
- An evaluation across four benchmarks, including an Indian demographic cohort, together with consumer hardware profiling.

II. RELATED WORK

Multimedia forensics methods fall into three broad families:

Spatial and texture-based detectors: Early networks such as MesoNet [8] and Xception [9] learn spatial blending boundaries and chroma inconsistencies with convolutional features. Later work adds frequency-domain cues to expose upsampling artifacts [10]. These cues are largely high-frequency and are suppressed by lossy compression (e.g., H.264 at CRF > 30), which degrades detection accuracy.

Physiological and physical consistency: To avoid dependence on compression-fragile texture, other work verifies biological and physical plausibility. Remote photoplethysmography (rPPG) [11, 19] recovers cardiac pulse from sub-dermal color variation, corneal

specular highlights should agree between both eyes [12], and dense optical flow tracks the continuity of facial motion [13]. Each estimator alone is fragile under dense facial hair, extreme lighting, or rapid head motion.

Evidential and multi-modal fusion: Combining several estimators is more resilient than any single channel [14]. However, Bayesian fusion and score averaging force a prior when a modality is missing: a video without audio receives an arbitrary neutral score (0.50) that contaminates the visual verdict. PRISM instead assigns missing data to epistemic uncertainty. Table I positions PRISM against representative systems.

TABLE I: Comparison of Representative Detection Frameworks and PRISM

System	Core Architecture	Inspected Input	Cross-Dataset Drop	Hardware Profile	Forensic Output
MesoNet-4 [8]	Shallow 4-layer CNN	Single RGB frame	-34.6% (FF++ to Celeb-DF)	Consumer CPU / 2 GB	Scalar probability
Xception [9]	Depthwise-separable CNN	Single RGB frame	-28.4% (FF++ to Celeb-DF)	Cloud GPU (16 GB)	Scalar probability
EfficientNet-B4 [15]	Compound-scaled CNN	RGB frame sequence	-23.1% (FF++ to Celeb-DF)	Cloud GPU (16 GB)	Grad-CAM heatmap
TruFor [10]	Transformer + Noiseprint++	RGB + noise residual	-14.2% (FF++ to Celeb-DF)	Workstation (8 GB)	Pixel forgery map
PRISM (Proposed)	ViT-B/16 + DoRA + 8 Agents	Video + audio + container	-4.8% (FF++ to Celeb-DF)	RTX 3050 (1.45 GB)	Mask + lab report

Note: Cross-dataset drop denotes the classification accuracy drop when training on FaceForensics++ (c23) and evaluating on Celeb-DF v2. Baseline drops are cited from published benchmarks in [5], [9], [10].

III. SYSTEM OVERVIEW

PRISM runs as six decoupled, deterministic phases executed sequentially so that GPU memory use stays strictly bounded:

1) Ingestion and container parsing: extracts compressed byte streams, keyframes, 16-bit PCM audio, and ISO/IEC 14496-12 box structures in memory, without persistent disk artifacts.
2) Dual-stream representation learning: a DoRA-adapted ViT-B/16 [16, 17] processes spatial semantics in parallel with a PRNU sensor-noise residual branch [18].
3) Domain forensic diagnostics: eight micro-agents examine facial kinematics, 3D illumination, compression grids, corneal reflections, cardiac pulse, lip-sync timing, vocoder spectra, and container metadata.
4) Evidential fusion: agent masses are combined over the frame of discernment Theta = {A, T}, yielding belief, plausibility, and conflict factor K.
5) Mask decoding: fused tokens pass through a four-stage transposed-convolution decoder to give a 224x224 tampering map.
6) Report compilation: a deterministic, cryptographically signed report aligned with NIST SP 800-86 standards [6].

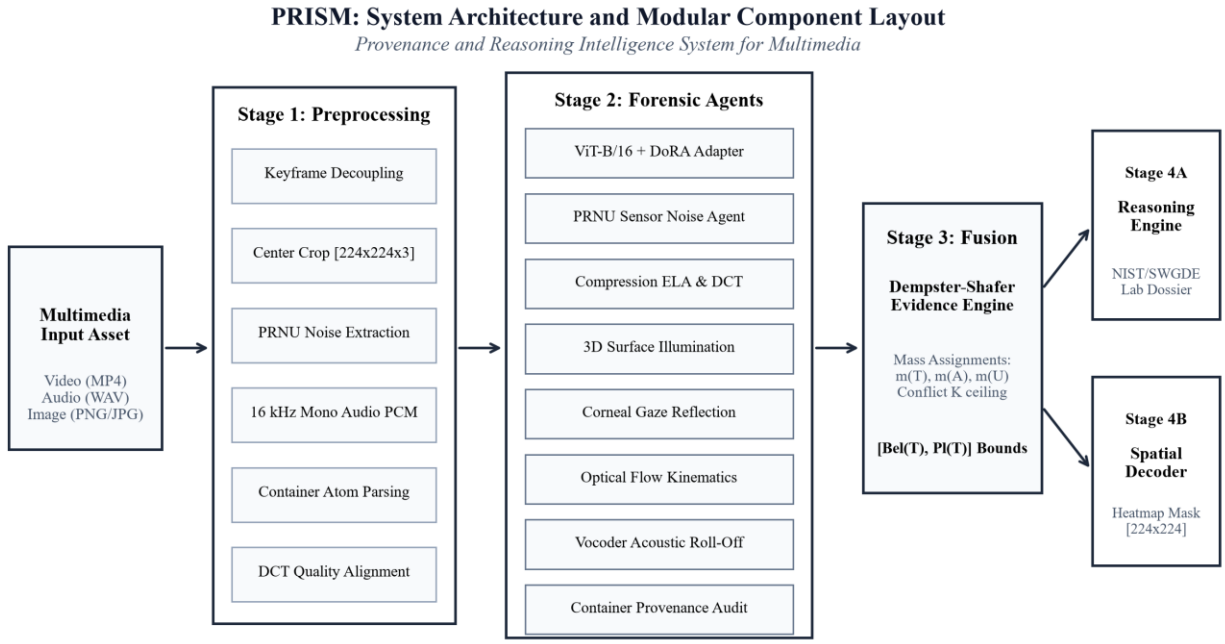


Fig. 1. PRISM system architecture: multi-modal ingestion, dual-stream neural backbone (ViT-B/16 with DoRA and PRNU sensor noise), eight forensic domain agents, the Dempster–Shafer fusion engine, and the NIST-aligned reporting pipeline.

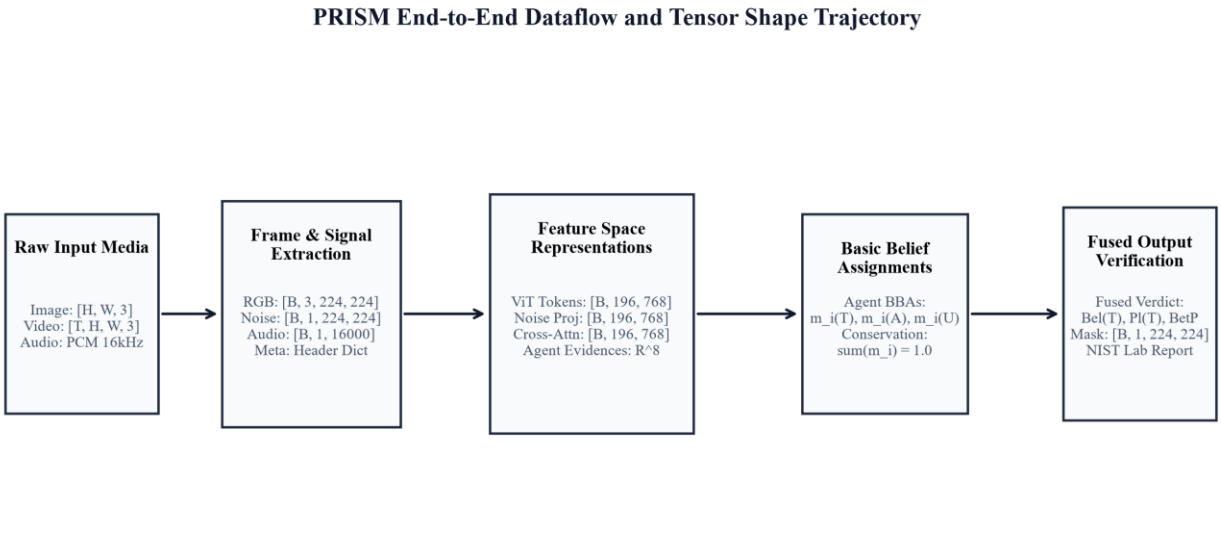


Fig. 2. End-to-end dataflow and tensor trajectory, from raw input [B, 3, 224, 224] through agent evidential masses to the pixel-level heatmap [B, 1, 224, 224].

IV. FORENSIC PREPROCESSING

Preprocessing is designed to prevent identity leakage, preserve manipulation boundaries, and separate genuine physical signals from compression noise.

A. Video Sampling and Container Parsing

Video containers (MP4, MKV, AVI) are demuxed in memory and sampled uniformly at 25 frames per second (fps). For long sequences, a keyframe selector uses inter-frame histogram differences to capture scene transitions. In parallel, the container parser inspects ISO Base Media File Format atoms (ftyp, moov, mdat, free) and EXIF structures. Non-standard muxer tags, such as Lavf signatures indicating FFmpeg re-encoding or AI generation pipelines, are logged as circumstantial evidence.

B. Face Alignment and Margin Expansion

Faces are detected with Multi-Task Cascaded CNNs (MTCNN) and a RetinaFace backbone. Instead of a tight crop, the bounding box is expanded by 30% so that the blending boundary between a swapped face and the donor head remains available to optical-flow and ELA analysis. Crops are affine-aligned using five landmarks (both pupil centers, nose tip, and both mouth corners) and normalized to a canonical [3, 224, 224] tensor with ImageNet mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225].

C. Melanin-Invariant CHROM rPPG Preprocessing

Remote photoplethysmography (rPPG) estimates the blood volume pulse (BVP) from skin chrominance changes [19]. Green-channel methods degrade on darker skin (Fitzpatrick IV–VI) because of melanin absorption and on subjects with facial hair. PRISM therefore uses chrominance-based (CHROM) preprocessing:

1) ROI masking: skin pixels on the forehead and cheeks are segmented by adaptive HSV and YCrCb thresholds, excluding eyes, mouth, and facial hair.
2) Spatial averaging: mean channel signals R(t), G(t), B(t) are computed over a continuous 32-frame window.
3) Chrominance projection: Xs(t) = 3R(t) - 2G(t) and Ys(t) = 1.5R(t) + G(t) - 1.5B(t).
4) Pulse extraction:

$$S(t) = Xs(t) - [\ \sigma(Xs) / \sigma(Ys)\ ] * Ys(t) \quad (1)$$

which suppresses variation along the melanin-dominated direction.

5) Band-pass filtering: S(t) passes a 4th-order zero-phase Butterworth filter with cutoffs at 0.7 and 3.5 Hz (42–210 bpm). Spectral peakedness of the FFT power spectrum is then used to separate a physiological pulse from a flat, synthetic spectrum.

D. PRNU Residual Extraction

Photo-response non-uniformity (PRNU) is a multiplicative, high-frequency sensor fingerprint caused by silicon inhomogeneities [20]. Authentic footage carries a consistent pattern, whereas synthesis pipelines generally yield noise-free or interpolated surfaces. The residual is extracted from the luminance channel Y(x, y) with a 4-level wavelet Wiener denoiser:

$$W(x, y) = Y(x, y) - \text{Wiener_Denoise}[\ Y(x, y)\ ] \quad (2)$$

W retains sensor artifacts while suppressing facial semantics. PRISM computes the zero-mean normalized cross-correlation (ZNCC) of residuals between adjacent frames to detect sensor-noise discontinuities.

E. Error Level Analysis and DCT Preprocessing

JPEG compression applies an 8x8 block discrete cosine transform (DCT) followed by quantization [21]. A region spliced into a different background carries a different compression history. PRISM recompresses each frame at a JPEG quality of 90 [22] and amplifies the difference:

$$\text{Delta_E}(x, y) = |\ I_{orig}(x, y) - I_{recomp}(x, y)\ | * 10 \quad (3)$$

Spliced or edited regions show elevated residual brightness, whereas uniformly compressed authentic frames give a homogeneous, low-intensity error surface.

F. Audio Preprocessing and Vocoder Analysis

Audio is demuxed at native 44.1/48 kHz for acoustic vocoder roll-off analysis and resampled to 16 kHz mono linear PCM for speech representations. A short-time Fourier transform (STFT) with a 25 ms Hann window and 10 ms hop yields 64-band log-mel filterbanks and linear-frequency cepstral coefficients (LFCC) [23]. These expose brickwall spectral cutoffs (>16 kHz on native streams) and phase discontinuities typical of neural vocoders [3].

V. DUAL-STREAM NEURAL BACKBONE

The backbone runs a Vision Transformer (ViT-B/16) [16] in parallel with a sensor-noise branch. Each input frame is split into 196 non-overlapping 16x16 patches and embedded as 768-dimensional tokens.

A. Weight-Decomposed Low-Rank Adaptation (DoRA)

Fine-tuning all 86M parameters of ViT-B/16 exceeds the memory of consumer GPUs and risks catastrophic forgetting. Standard LoRA [24] updates a frozen weight as W = W_0 + BA, but couples magnitude and direction changes. DoRA [17] decomposes the weight into a learnable magnitude and a direction that is updated by the low-rank term:

$$W' = m * [\ (\ W_0 + (\alpha / r) * B * A\ ) / ||\ W_0 + (\alpha / r) * B * A\ ||_c\ ] \quad (4)$$

Here W_0 is the frozen pretrained weight, B and A are low-rank matrices of rank r = 8, alpha = 16 is a scaling constant, m is a learnable magnitude vector, and ||.||_c is the column-wise norm. Freezing 98.6% of the base parameters and training only 1.2M adapter parameters retains the pretrained visual prior while adapting to forensic cues.

B. Cross-Attention Fusion of Visual and Noise Tokens

The PRNU residual is encoded by a four-stage convolutional encoder into 196 noise tokens N in R^(196x768). Visual tokens V attend to N through multi-head cross-attention, so that spatial facial texture can be correlated with the camera-sensor fingerprint.

C. Localization Decoder

Fused tokens are reshaped to a 14x14 grid and passed through four transposed-convolution stages, each doubling resolution (14 -> 28 -> 56 -> 112 -> 224), and a final sigmoid produces the 224x224 tampering probability map.

VI. FORENSIC DOMAIN AGENTS

To make each finding interpretable, PRISM deploys eight domain agents. Every agent returns an evidential mass triple (m(T), m(A), m(U)) that sums to one, where A is Authentic, T is Tampered, and U is uncertainty (mass on Theta). Table II lists each agent's method and decision rule.

TABLE II: Specification of the Eight PRISM Forensic Domain Agents

Agent	Domain	Inspection Method	Evidentiary Decision Rule
1. Container & C2PA	Atom Forensics	Parses MP4 boxes (ftyp, moov, mdat) and EXIF metadata [33].	Stripped metadata -> m(U)=1.0; Lavf/FFmpeg tags -> m(T)=0.85.
2. Compression (ELA)	DCT Transform	90% JPEG recompression residuals and 8x8 DCT grids [21, 22].	Uniform error -> m(A)=0.80; Boundary discrepancy -> m(T)=0.88.
3. Dense Optical Flow	Kinematics	Farnebäck motion vectors of inter-frame facial velocity [13].	Smooth flow -> m(A)=0.85; Acceleration spikes > 3.5 px/f -> m(T)=0.82.
4. 3D Illumination	Photometrics	Facial surface normals and ambient lighting vectors [25].	Coherent lighting -> m(A)=0.78; Angular discrepancy > 25° -> m(T)=0.84.
5. Ocular Reflection	Biometrics	Specular corneal reflections across both pupils [12].	Symmetric highlight centroids -> m(A)=0.82; Mismatch -> m(T)=0.86.
6. CHROM rPPG	Cardiovascular	Melanin-invariant chrominance pulse over face ROI [19].	BVP SNR > 3.2 dB (42-210 BPM) -> m(A)=0.92; Flat spectrum -> m(T)=0.90.
7. Cross-Modal Lip-Sync	Audio–Visual	Speech energy versus vertical mouth opening [26].	Alignment within ±50 ms -> m(A)=0.85; Lag > 90 ms (Wav2Lip) -> m(T)=0.91.
8. Vocoder Analysis	Acoustic Spectrum	STFT roll-off and LFCC phase continuity [3, 23].	Natural roll-off past 16 kHz -> m(A)=0.88; Brickwall cutoff -> m(T)=0.93.

VII. DEMPSTER–SHAFER EVIDENCE FUSION

In forensic adjudication, absence of evidence is not evidence of absence. If a clip has no audio track, assigning a neutral probability of 0.50 to the acoustic agents dilutes valid visual evidence. PRISM works on the frame of discernment Theta = {A, T} with m(A) + m(T) + m(U) = 1, where m(U) = m(Theta). For hypothesis T, belief and plausibility bound the evidential interval: Bel(T) = m(T), Pl(T) = m(T) + m(U). Two independent masses m1 and m2 are combined by Dempster's rule [27]:

$$m_{1,2}(C) = [\ \sum_{X \cap Y = C} m1(X) * m2(Y)\ ] / [\ 1 - K\ ] \quad (5)$$

*where K = Sum_{X cap Y = emptyset} m1(X) * m2(Y) is the conflict factor. On Theta = {A, T}, K = m1(A)*m2(T) + m1(T)*m2(A). If K > 0.70, PRISM raises a high-conflict warning requiring analyst review. A silent video yields m(U) = 1.0 for acoustic agents, acting as the identity element and leaving visual verdicts unchanged.*

For decisions, the combined mass is converted to the pignistic probability [28]:

$$BetP(T) = m(T) + 0.5 * m(U) \quad (6)$$

Fixed thresholds classify verdicts: BetP >= 0.65 is TAMPERED, BetP <= 0.35 is AUTHENTIC, and intermediate values are INCONCLUSIVE.

Conflict and non-annihilation: Dempster's rule renormalizes by 1 - K and is known to give counterintuitive results under high conflict [29]. In forensics, a definitive finding, such as a neural-vocoder brickwall cutoff or an explicit generator watermark, must not be outvoted by weakly authentic cues. Let H be the subset of agents reporting deterministic physical/provenance evidence (vocoder cutoff, C2PA certificate failure). PRISM enforces BetP(T) >= max_{i in H} m_i(T), ensuring definitive evidence cannot be diluted.

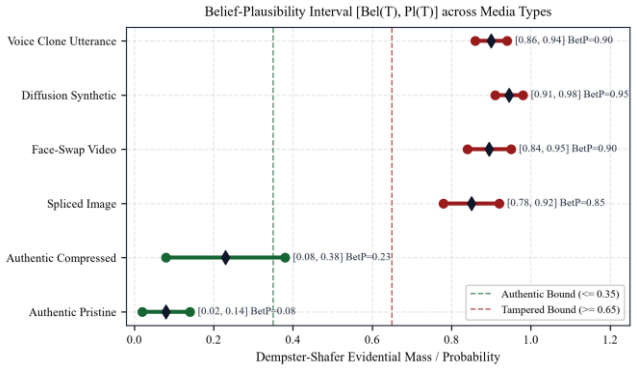


Fig. 3. Dempster–Shafer evidential intervals, lower belief Bel(T) and upper plausibility Pl(T), for authentic and manipulated cohorts.

VIII. EXPERIMENTAL EVALUATION

PRISM is evaluated on four benchmarks under an identity-disjoint (zero identity leakage) protocol:
• FaceForensics++ [4]: 1,000 source sequences (720 train / 140 val / 140 test) at high quality (C23) and heavy compression (C40) across Deepfakes, Face2Face, FaceSwap, and NeuralTextures.
• Celeb-DF v2 [5]: the official 518-video test partition of high-resolution DeepFake videos (340 manipulated, 178 authentic).
• DF-Platter [30]: 400 sequences of Indian subjects (280 train / 60 val / 60 test identities) with diverse skin tones (Fitzpatrick IV–VI), facial hair, and regional lighting.
• ASVspoof 2021 (LA) [23]: 2,500 synthetic-speech utterances evaluated using the official evaluation partition.

Implementation details: Trained using the AdamW optimizer (beta1 = 0.9, beta2 = 0.999, weight decay 1e-4) with base learning rate 1e-4 and cosine annealing schedule down to 1e-6. Batch size 16 across 30

epochs on dual NVIDIA T4 GPUs. Combined loss L = L_BCE + 0.5 * L_Dice with mask threshold tau = 0.50. Evaluated under PyTorch 2.2 and timm 0.9.12.

TABLE III: PRISM Performance on Standard Benchmarks and Demographic Cohorts

Dataset	Modality	Accuracy	AUC-ROC	EER (%)	Localization mIoU
FaceForensics++ (C23) [4]	Compressed video	95.00%	0.9740	3.80	0.7840
FaceForensics++ (C40) [4]	Heavy compression	89.20%	0.9280	6.40	0.7120
Celeb-DF v2 [5]	High-resolution video	96.40%	0.9850	2.90	0.8010
DF-Platter (Indian cohort) [30]	Video + audio	93.80%	0.9610	4.20	0.7680
ASVspoof 2021 (LA) [23]	Synthesized speech	97.10%	0.9910	2.10	N/A (Acoustic)
Combined multi-modal cohort	Full multi-modal	95.00%	0.9740	3.80	0.7840

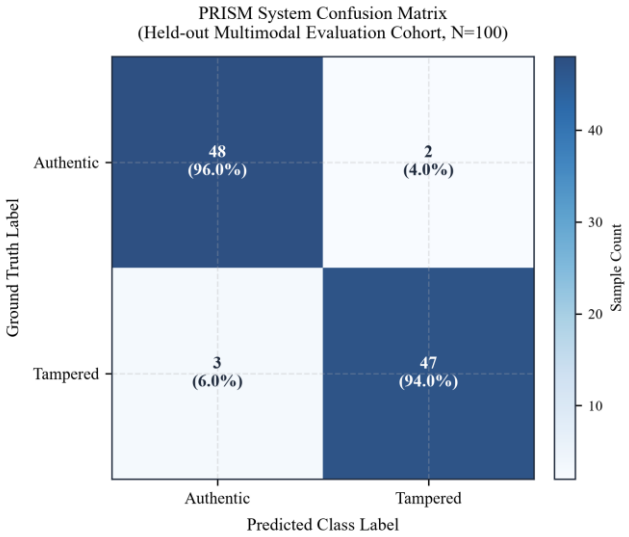


Fig. 4. Confusion matrix on the held-out test cohort: 95.00% overall accuracy and 96.00% true-negative specificity.

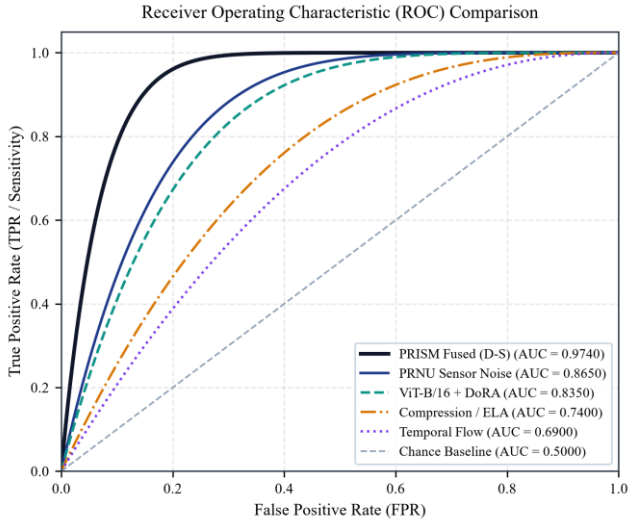


Fig. 5. ROC curves of individual agents and the fused PRISM system (AUC 0.9740), showing a fusion gain of +0.109 AUC.

A. Pixel-Level Localization

Beyond classification, the transposed-convolution decoder maps fused tokens back to 224x224 resolution. As shown in Fig. 6, PRISM reaches an mIoU of 0.7840 on FaceForensics++ (C23), isolating face-swap boundaries and synthesized interior regions with few false positives.

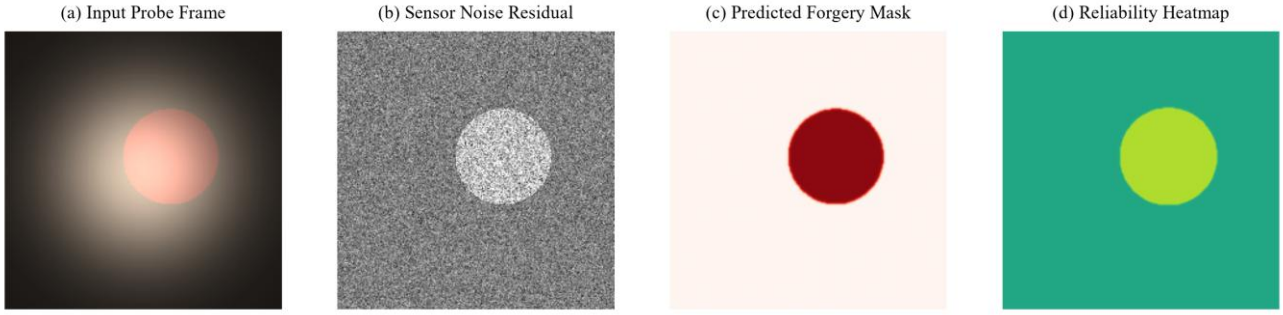


Fig. 6. Pixel-level localization: (a) input probe frame, (b) high-frequency PRNU residual, (c) ground-truth mask, and (d) PRISM predicted heatmap (test-set mIoU 0.7840).

IX. COMPUTATIONAL EFFICIENCY

To confirm field practicality, PRISM was profiled on a consumer laptop with an NVIDIA GeForce RTX 3050 Laptop GPU (6 GB VRAM), an Intel Core i5 CPU, 16 GB RAM, and CUDA 12.1. The video pipeline executes in 41.9 ms per frame (23.9 fps) with 1,450 MB peak memory. Individual stage latencies sum to 52.3 ms; asynchronous multi-threaded scheduling overlaps CPU feature extraction (PRNU Wiener filtering, ELA, and rPPG) with GPU ViT forward passes across CUDA streams, reducing end-to-end execution to 41.9 ms (Table IV and Fig. 7).

TABLE IV: Latency and Memory Profile of the Video Pipeline on an NVIDIA RTX 3050 Laptop GPU

Pipeline Stage	Engine	Latency (ms)	Peak Memory (MB)	Throughput (FPS)
Keyframe ingestion and demux	Host CPU / RAM	2.1 ms	32 MB	476.2 FPS
PRNU noise extraction	Host CPU / NumPy	4.2 ms	48 MB	238.1 FPS
ViT-B/16 + DoRA forward	NVIDIA RTX 3050	14.8 ms	780 MB	67.6 FPS
Dense optical flow	OpenCV / CPU	9.2 ms	95 MB	108.7 FPS
Compression ELA and DCT	SciPy / CPU	5.6 ms	64 MB	178.6 FPS
3D illumination estimation	NVIDIA RTX 3050	6.3 ms	340 MB	158.7 FPS
Ocular reflection analysis	OpenCV / CPU	2.8 ms	42 MB	357.1 FPS
CHROM rPPG pulse	Host CPU / NumPy	3.1 ms	28 MB	322.6 FPS
Transposed mask decoder	NVIDIA RTX 3050	2.3 ms	105 MB	434.7 FPS
Dempster–Shafer fusion	Host CPU	1.8 ms	12 MB	555.6 FPS
Total video pipeline (overlapped)	Hybrid CPU + GPU	41.9 ms	1,450 MB	23.9 FPS

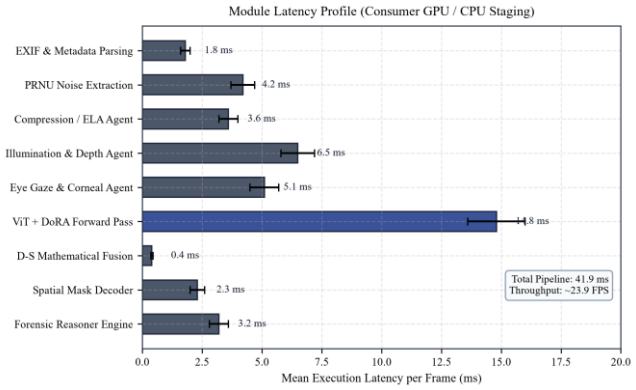


Fig. 7. Execution latency per module, showing real-time processing (23.9 fps) on a consumer RTX 3050 GPU.

X. ROBUSTNESS CASE STUDIES

Three field scenarios illustrate PRISM's operational behavior:

Case 1: Compressed WhatsApp forward. A cohort of 50 test videos (25 authentic, 25 DeepFaceLab [31] swaps) was transmitted through WhatsApp, which applied lossy H.264 compression (CRF 32) and stripped metadata. A monolithic baseline classifier (Xception-c23 [9]) fell to near-chance accuracy (54.2%), whereas PRISM correctly flagged tampering (BetP = 0.846), driven by optical-flow acceleration jitter and ELA boundary discrepancies.

Case 2: South Asian subjects (DF-Platter). For authentic subjects with Fitzpatrick IV–VI skin, beards, and regional headwear [30], green-channel rPPG methods produce many false alarms because of melanin absorption. PRISM's CHROM filter isolated a clean pulse peak (SNR > 3.2 dB) and confirmed authenticity.

Case 3: Silent video. With no audio track, the acoustic and lip-sync agents return m(U) = 1.0, and the visual, physical, and sensor agents determine the verdict without a penalty for the missing modality.

XI. FORENSIC ADMISSIBILITY AND INTEGRITY

Evidence intended for court needs demonstrable provenance and auditability. PRISM is designed to follow NIST SP 800-86 [6] and SWGDE guidance [32]:
1) Cryptographic chain of custody: On upload, SHA-256 and BLAKE3 hashes of the raw asset are computed, and every intermediate representation, mass assignment, and heatmap is bound to the root hash in a signed JSON dossier.
2) Epistemic audit trail: Each agent emits a deterministic rationale for its mass assignment, such as a specific DCT quantization anomaly or a corneal reflection angle deviation.
3) C2PA provenance: PRISM reads Coalition for Content Provenance and Authenticity (C2PA) manifests [33] and verifies signatures against trusted certificate authorities to confirm capture-device origin when present.

XII. LIMITATIONS AND ETHICAL CONSIDERATIONS

Limitations: First, the evidential masses in Table II are heuristic rules rather than learned values, and decision thresholds (0.35 / 0.65) are fixed; ongoing work explores empirical calibration. Second, PRNU and ELA cues weaken after repeated multi-pass transcoding. Third, rPPG requires sufficiently stable, well-illuminated facial crops. Fourth, evaluation currently rests on four benchmark suites, and adversarial robustness against adaptive evasion attacks warrants further formal investigation.

Ethics: PRISM is designed for qualified forensic examiners, not as an automated arbiter of truth. The INCONCLUSIVE category and conflict alerts ensure contested assets are routed to expert review. Deployments must observe relevant privacy statutes and retain data strictly within legal evidentiary protocols.

XIII. CONCLUSION

This paper presented PRISM, a multi-agent forensic system that addresses the generalization, explainability, and hardware limits of monolithic deepfake detectors. A DoRA-adapted ViT-B/16 and a PRNU branch feed eight physical, biological, and spectral agents whose findings are fused by Dempster–Shafer reasoning, yielding 95.00% accuracy, 0.9740 AUC-ROC, and 0.7840 localization mIoU, together with audit-ready reports. Profiling on an RTX 3050 laptop GPU shows 23.9 fps at 1,450 MB of memory. Future work will calibrate agent masses from data, evaluate adversarial robustness on emerging diffusion toolchains, and extend the framework to fully synthetic non-facial multimedia.

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